University of Engineering & Technology Peshawar

Computer Programming (C++) Lab

Lab 08

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DATE:	29/04/26
LAB TITLE:	TWO DIMENSIONAL ARRAY IN C++

SUBMITTED TO ENGR RIZWAN ULLAH

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Learning Objectives:

The objective of this lab is to understand and apply two-dimensional arrays in C++. Students will learn how to:

- Declare and initialize 2D arrays in C++.
- Access and display elements of a 2D array using nested loops.
- Perform row-wise and column-wise sum calculations on 2D arrays.
- Find the maximum value in a 2D array.

Introduction:

The simplest form of a multidimensional array is the two-dimensional (2D) array. A two-dimensional array is essentially a list of one-dimensional arrays arranged in rows and columns, similar to a matrix or table.

To declare a 2D array:

Syntax:

```
type arrayName[rows][columns];
```

Example: int matrix[3][4]; declares a 2D integer array with 3 rows and 4 columns.

To initialize a 2D array:

```
int a[3][4] = {
    {0, 1, 2, 3},   // row 0
    {4, 5, 6, 7},   // row 1
    {8, 9, 10, 11}  // row 2
};
```

Elements are accessed using two indices: a[row][column]. Nested for loops are typically used to iterate over all elements.

Task 1: Two-Dimensional Array Declaration and Initialization

Declare a 3x3 integer array called 'matrix', initialize it with values 1 through 9, and display all elements using nested loops.

Code:

```
#include <iostream>
using namespace std;

int main()
{
    // 3x3 ka 2D array declare aur initialize kar rahe hain
    int matrix[3][3] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

    cout << "Matrix ke elements hain:\n";

    // Nested loop se matrix ke tamam elements display karenge
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            // Har element print karna
```

```
            cout << matrix[i][j] << " ";
        }

        // Har row ke baad next line
        cout << endl;
    }

    return 0;
}
```

Output:

```
Matrix ke elements hain:
1 2 3
4 5 6
7 8 9
```

Task 2: Sum of Rows and Columns in a Two-Dimensional Array

Declare and initialize a 3x3 integer array with random values. Calculate and display the sum of each row, each column, and the total sum of all elements.

Code:

```
#include <iostream>
using namespace std;

int main()
{
    // 3x3 array declare aur initialize
    int matrix[3][3] = {
        {2, 4, 6},
        {1, 3, 5},
        {7, 8, 9}
    };

    int totalSum = 0;

    // Rows ka sum nikalna
    cout << "Rows ka sum:\n";

    for (int i = 0; i < 3; i++)
    {
        int rowSum = 0;

        for (int j = 0; j < 3; j++)
        {
            rowSum = rowSum + matrix[i][j];
            totalSum = totalSum + matrix[i][j];
        }

        cout << "Row " << i + 1 << " ka sum = " << rowSum << endl;
    }

    cout << endl;

    // Columns ka sum nikalna
    cout << "Columns ka sum:\n";

    for (int j = 0; j < 3; j++)
    {
        int colSum = 0;

        for (int i = 0; i < 3; i++)
        {
            colSum = colSum + matrix[i][j];
```

```
        }

        cout << "Column " << j + 1 << " ka sum = " << colSum << endl;
    }

    cout << endl;

    // Total sum display karna
    cout << "Tamam elements ka total sum = " << totalSum << endl;

    return 0;
}
```

Output:

```
Rows ka sum:
Row 1 ka sum = 12
Row 2 ka sum = 9
Row 3 ka sum = 24

Columns ka sum:
Column 1 ka sum = 10
Column 2 ka sum = 15
Column 3 ka sum = 20

Tamam elements ka total sum = 45
```

Task 3: Finding the Maximum Value in a Two-Dimensional Array

Declare and initialize a 4x4 integer array with random values. Find and display the maximum value in the entire array.

Code:

```
#include <iostream>
using namespace std;

int main()
{
    // 4x4 array declare aur initialize
    int matrix[4][4] = {
        {12, 25, 7, 18},
        {30, 14, 9, 22},
        {5, 17, 40, 11},
        {8, 19, 13, 27}
    };

    // Pehla element maximum maan lete hain
    int max = matrix[0][0];

    // Nested loop se maximum value find karenge
    for (int i = 0; i < 4; i++)
    {
        for (int j = 0; j < 4; j++)
        {
            // Agar current element bara ho to max update karo
            if (matrix[i][j] > max)
            {
                max = matrix[i][j];
            }
        }
    }

    // Maximum value display karna
    cout << "Matrix ki maximum value = " << max << endl;
```

```
    return 0;
}
```

Output:

```
Matrix ki maximum value = 40
```